

20250623EB_MN (ATP, NCT, TDP43)

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- 2025年6月23日 星期一
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2025年6月25日 星期三
- T2 -> T3
- 2025年6月27日 星期五
- T3 -> T4 (~14 mL for over-weekend)
- 2025年6月30日 星期一
- T4 -> T5
- 2025年7月2日 星期三
- T5 -> T6
- 2025年7月4日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2025年7月7日 星期一
- T6 -> T7
Dissociation
 - Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
 - Remove the PLO and wash the plate with PBS once
 - Coat the plate with laminine for 1hr.
 - Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
 - Add 2mL accutase in 15mL falcon
 - Put the tube into 37C water bath. Flick the cells every 5 min until the accutase went cloudy (~45min this time). (There are many cells attached on the well of 15mL falcon. Will use 50mL next time)
 - pipette the accutase gently and slowly around 20-30 times to break the EBs by checking if there are still visible EBs. (I can still see a lot of undissociated EBs this time, so I take the accutase without visible EBs for the further process.
 - Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.

9. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
10. Filter the cell with 40um filter into a new 50mL tube
11. Spin the cell down at 100rcf, 3.5min. Remove the supernatent.
12. Add 1ml of T7 to resuspend the cells thouroughly to avoid clusters.
13. Add more T7 (~5mL) to resuspend the cells and count it.
14. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

coating date ^

	A	B	C
1		A	B
2	0.01% PLO with Na borate	2025/6/20	2025/6/20
3	laminin	2025/6/23	2025/6/23

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	P1	5

Table1 ^

	A	B
1	run1	volume (mL)
2	accutase	2
3	HI-FBS/CTWM	4

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	7.65	97	3	22.95	A,B	24	250	32	1046

*Use new 110 and 111 virus

Plate A							^	
	1	2	3	4	5	6		
A	92 5uL in total	110 12uL in total	111 12uL in total	no virus				
B								
C								
D								

Plate B (for Stephanie)							^
	1	2	3	4	5	6	
A	92 5uL in total	129 20uL in total	74 20uL in total				
B							
C							
D							

- 2025年7月8日 星期二
- Change T7 -> T7
- 2025年7月10日 星期四
- T7 -> BP T8 (with 1mM extra pyruvate)
- 2025年7月11日 星期五
- 1/2 T8 -> 1/2 T8
- 2025年7月14日 星期一
- 1/2 T8 -> 1/2 T8
- 2025年7月16日 星期三
- 1/2 T8 -> 1/2 T8
- 2025年7月17日 星期四
- 2025年7月18日 星期五
- Image Plate A

Plate A conditions							^
	1	2	3	4	5	6	
A	1	1	1	1	1		
B	2	2	2	2	2		
C	3	3	3	5	5		
D	4	4	4	1_24h	2_24h		

condition from 20250527MN1														✓
	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	media	pyruvate (100x)	100mM pyruvate	N2(100x), B27 (50x)	N2	B27	GlutaMax (100x)	Glutamax	T8 supp	glucose	1.1M glucose	drug	volume (uL)	
2	NB-A	1mM	300	v	300	600	v	300	v	0			30000	
3	NB-A	1mM												

media prep											✓
	A	B	C	D	E	F	G	H	I	J	
1											
2	condition #	glucose conc	diluted from (mM)	glucose volume	total volume (uL)	drug	volume w/ SirDNA	SirDNA (1:5000)	volume w/ JFX650 & SPY595	JFX650 & SPY595 (1:2500)	
3	1	2.5	1100	12.50	5500		1600	0.32	1600	0.64	
4	2	0.15	2.5	300.00	5000		1600	0.32	1600	0.64	
5	3	0		0	2500		1600	0.32			
6	4	0		0	2500	FCCP (3uL), 2DG (9uL)	1600	0.32			
7	5	2.5			800	2uM MG132			800	0.32	
8	1	2.5	1100	1.82	800				800	0.32	
9	2	0.15	2.5	48.00	800				800	0.32	
10	5	0			800	2uM MG132			800	0.32	

MG132
stock: 10.5mM
working: 2uM
dilution: 5250
dilute 1 to 10 first -> add 1.52 uL in 800uL

Treatment time (all)												
	A	B	C	D	E	F	G	H	I	J	K	L
1			110 (plate A)		111 (plate A)		92 (plate A)		TDP43 (plate A)			
2	#		treat time	image time	treat time	image time	treat time	image time	treat time	treat JFX, SPY	image time	
3	1	2.5mM glu	12:30	13:30	9:50	10:58						
4	2	0.15mM glu	13:10	13:37	10:20	11:30						
5	3	0mM glu	13:50	15:03	10:53	12:04						
6	4	0mM glu+FCCP+ 2DG	14:55	15:59	11:28	12:33						
7	1	2.5mM glu					15:25	16:34				
8	2	0.15mM glu					15:40	16:42				
9	3	0mM glu					15:55	16:57				
10	4	0mM glu+FCCP+ 2DG					16:20	17:23				
11	1	2.5mM glu								16:45	17:47	
12	2	0.15mM glu								17:30	18:45	
13	5	2.5mM glu +2uM MG132 6h							12:00	17:00	18:10	
14	1	2.5mM glu								17:50	18:58	
15	2	0.15mM glu								18:10	19:10	
16	5	2.5mM glu +2uM MG132 6h							12:00	17:15	18:15	
17	7/19											
18	1_24h	2.5mM glu							7/18 19:10			
19	2_24h	0.15mM glu							7/18 19:10			

Image (NCT, column 2 and 3)

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	



8 positions/well (10 positions sometimes delayed the imaging a bit)

*111 didn't express well. Need to add more virus. The mCherry+ cells are okay, but the intensity can be improved.

111_4 position 2 didn't look well. Most of them don't have good expression

Also, cells clustered too much, so it's a bit hard to do imaging

*110_4 out of focus the first time. Re-defined all the positions, so start the acquisition ~30 min later than the time point

*110_6: position 7 is out of focus due to bubbles from frame 12-18; position 8 is out of focus from frame 11-17.

Image (ATP/ADP ratio with vLH92, column1)

O.C. , 60X, 600x900, 16-bit				
	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	20	200
3	Dapi-ex-Green-em	16	20	300



10positions/well

Treat cells for an hour and image the end point ATP/ADP ratio

*condition3 looked weird. There are lots of empty holes between neurites and they looked disturbed. Can't find any non-MNs

*try to find non-MN cells: can't find non-MNs in condition 3 and 4

Image (TDP43, column 4 and 5)

Treat cells with 2.5uM JFX650 and SPY595 (1:2500) 1h before the imaging. (0.16uL per well)

dilute 1mM JFX650

--> 1:400 --> 2.5uM

--> 1:2500 --> 1nM (working concentration)

O.C. , 60X, 600x900, 16-bit1						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	561	16	20	500ms		
3	640	16	10	300ms		
4	640 movie	16	40	500ms	40 loops	

Imaging:

~10 positions for snapshot (odd fileids)

6 positions for neurites (even fileids)

Table2								
	A	B	C	D	E	F	G	H
1	fileid	treatment	treat time	Halodye time	image type	well		
2	1	2.5mM full supp	1h	1h Halodye	snapshot	4A		
3	2	2.5mM full supp	1h	1h Halodye	movie			
4	3	2.5mM full supp + 2uM MG132	6h	1h Halodye	snapshot	4C		
5	4	2.5mM full supp + 2uM MG132	6h	1h Halodye	movie			
6	5	2.5mM py only + 2uM MG132	6h	1h Halodye	snapshot	5C		
7	6	2.5mM py only + 2uM MG133	6h	1h Halodye	movie			
8	7	0.15mM full supp	1h	1h Halodye	snapshot	4B		
9	8	0.15mM full supp	1h	1h Halodye	movie			
10	9	2.5mM py only	1h	1h Halodye	snapshot	5A		
11	10	2.5mM py only	1h	1h Halodye	movie			
12	11	0.15mM py only	1h	1h Halodye	snapshot	5B		
13	12	0.15mM py only	1h	1h Halodye	movie			
14	13	2.5mM full supp	24h	without out new Halodye	snapshot	4A	image at 9:10-9:25	
15	14	0.15mM full supp	24h	without out new Halodye	snapshot	4B		
16	15	2.5mM py only	24h	without out new Halodye	snapshot	5A		
17	16	0.15mM py only	24h	without out new Halodye	snapshot	5B		
18	17	2.5mM full supp	24h	1h Halodye (at 7/19 9pm)	snapshot	4D	image at 10:04	
19	18	0.15mM full supp	24h	1h Halodye (at 7/19 9pm)	snapshot	5D	image at 10:10	
20	19	2.5mM full supp	24h	1h Halodye (at 7/19 9:35pm)	snapshot	4A	image at 10:34	Use same positions as 13
21	20	0.15mM full supp	24h	1h Halodye (at 7/19 9:35pm)	snapshot	4B		
22	21	2.5mM py only	24h	1h Halodye (at 7/19 9:35pm)	snapshot	5A		
23	22	0.15mM py only	24h	1h Halodye (at 7/19 9:35pm)	snapshot	5B		

Change half media for 4A,B, D and 5A,B,D with their corresponding treatment

2025年7月23日 星期三

Image D5